Personal Reflection

The movie was a little confusing at times and I didn't understand the context of the future past memories. I had to watch the movie twice and then understood much better that time is not linear, in the movie.

As I watched, I was thinking of the many ways that NLP (and computer vision!) could be used to bridge the gap of being able to talk to another person/alien/lifeform when the languages are so completely different. There were several ambiguities that were evident in the film – from how to interpret the word "weapon" versus tool and other cultural differences that influence how context influenced perceived meaning. This is very similar to real-world issues in language translation and semantic understanding.

There were so many ways that I thought of tackling how to decipher the heptapods language from previous classes. Since there were two methods of communicating, I thought Word2Vec or Transformers for the ink/written communication and then audio embedding or feature extraction for the verbal communication. The part that would be most difficult is the translation of the vectors in human language. In the film, Louise became quickly fluent in heptapod – which would require far more time and computational effort in reality – but the thought process helped me think through how these methods might be applied.

This assignment gave me the chance to put all that I have learned in my classes to the test. While I have studied many techniques, they were taught as isolated weekly tasks. So, I had to go back and lookup what I had learned previously. I was frustrated in that I had forgotten a lot of previously learned material, and was also glad that I am applying what was learned.